1. What I built and why

Python five-stage research QA: route, retrieve with BM25, build grounded prompts, stream Groq output, and validate citations. A citation-repair branch is available. I prioritize grounded answers over accepting every request; paper scope must constrain retrieval. Ignoring it caused three wrongful Transformer refusals in the baseline.

2. The real service and my stand-in

Groq openai/gpt-oss-20b: 3 successful calibration calls gave model first-content median 884.57 ms and total median 983.85 ms; the sample cannot establish tail behavior. Earlier low-output-cap tests truncated. The FastAPI stand-in streams SSE and simulates quotas, concurrency, failures and slow tails from these 3 samples. Its quota/token model is approximate; lab TPM 50,000 differs from real 8,000.

Table 1. Real-service limits

Limit	Value	Evidence	Confidence
Requests/min	30	Groq documentation	medium
Tokens/min	8,000	Headers + docs	high
Requests/day	1,000	Headers + docs	high
Tokens/day	200,000	Groq docs only	medium
Max concurrency	not measured	No real saturation	low
Limit response	429 (docs)	Not observed live	medium
Recovery time	not measured	No live 429	low

3. Results

Five load shapes used stand-in only. Percentiles use successful requests. Controlled 30 vs 15 RPM (24 requests each): successes fell from 24 to 15, a 37.5 percentage-point drop. This isolates RPM better than the original confounded comparison.

Table 2a. Latency | p50 / p95 / p99 milliseconds

Scenario	Source	N	Duration (s)	Offered /min	First model content (ms)	Total latency (ms)
Steady	stand-in	8	18.0	24.0	904 / 961 / 969	998 / 1049 / 1050
Burst	stand-in	24	2.7	5971.4	1665 / 2452 / 2514	1761 / 2603 / 2679
Overload	stand-in	48	8.3	399.9	1671 / 4167 / 4256	1780 / 4408 / 4496
RPM halved	stand-in	24	3.6	999.2	1928 / 2567 / 2672	2011 / 2731 / 2834
Day exhausted	stand-in	12	1.6	665.9	1162 / 1302 / 1314	1249 / 1398 / 1411

Table 2b. Outcomes | timeout/gave-up includes timeout and error

Scenario	OK	Rate limited	Timeout / gave up	Retries / request	Peak queue	Tokens / OK answer	Cost / 1k
Steady	100.0%	0.0%	0.0%	0.00	0	1030.1	not measured
Burst	25.0%	0.0%	75.0%	0.00	12	1054.5	not measured
Overload	20.8%	0.0%	79.2%	0.00	7	1127.5	not measured
RPM halved	37.5%	0.0%	62.5%	0.00	12	952.2	not measured
Day exhausted	25.0%	75.0%	0.0%	0.00	0	1058.3	not measured

4. Where the time goes

Slow stand-in request 7e558f9d-4f34-4f7a-a749-bce4f5d6bed9: queue 958.15 ms; pre-first-content work 3320.47 ms; post-first-content work 239.11 ms; retries 0; total 4517.73 ms. Model content began at 4278.62 ms; buffered reply ready at 4517.70 ms. Work includes HTTP/application overhead.

5. When a limit is hit

Bounded FIFO admission reserves request/token capacity. At a minute limit, eligible callers queue until their deadline; excess/expired requests are rejected. At the daily limit, immediate rejection is preferable to waiting or a weaker, ungrounded answer. Recovery time is unknown on the real service.

Retrieval: quality versus cost

The same 15 questions were run under three settings on real Groq (45 calls): 6 existing human decisions, 39 offline rubric decisions. The rubric checks core concepts and citation provenance, not every claim. Balanced is preferred for its less extreme observed latency tail; small saves tokens near TPM pressure (manual choice, not automatic).

Table 3. Real Groq retrieval quality | rubric-screened grades; provider first content

Setting: words/top-k/rerank	N	Correct + cited*	Wrong incl. false refusal	Proper refusal	Tokens / answer	First content p50/p95 (ms)	Total p50/p95 (ms)
small: 250w / 2 / no	15	73.3%	6.7%	20.0%	1110.9	1310 / 13617	1359 / 14033
balanced: 500w / 4 / no	15	73.3%	6.7%	20.0%	2879.6	1133 / 5625	1547 / 6300
reranked: 500w / 4 / lexical	15	73.3%	6.7%	20.0%	3085.3	1225 / 5161	1588 / 5520

How this library will mislead a naive retriever

(1) Shared architecture terms: the Transformer encoder/decoder question retrieved H-Net and caused three refusals. (2) H-Net figures contain garbled extracted glyphs, so chunk-boundary queries rank noisy evidence. (3) An unanswerable stock-price query retrieved Moshi through incidental word overlap.

6. Retries

For transient failures, respect Retry-After and the caller's deadline. Never retry daily quota exhaustion or a partial streamed response. In a simulated 1-RPM test, naive retries added 756.83 ms and two attempts with no recovered answer. The retry module was tested separately, not wired into normal requests.

7. What the person sees

At 3pm with no daily budget: 'Today's research-answer budget is exhausted. I cannot verify an answer now and will not guess. Please try after the quota resets.' Proposed wording, not a tested UI.

8. What I'm unsure about, what I left out, what I learned

Three successful real latency samples cannot characterize burst tails; the stand-in depends on that assumption. I deliberately omitted a vector database and production UI. I learned that retry budgets must account for user patience, not only HTTP statuses.

Your feedback on Vaani

Website-only impression (verify personally): multilingual voice and turn-taking stood out; different latency claims were confusing. I would define each timer's start and stop beside the claim. I did not test the product.

How I worked with the assistant

Three prompts mattered: ground answers in exactly three papers (avoid guessing); provide complete copy-paste scripts/tests (reproducibility); assess answers rather than require 45 manual grades (reduce unnecessary work). I identified citation parsing failures from traces and overruled manual per-answer grading. I ran and diagnosed the experiments; the code was AI-assisted. Attach the unedited chat history separately.